HR Analytics – Salary Prediction

Business Report

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Business Context:

No one wants to feel undervalued when it comes to the compensation they receive for their skills and contributions, right? Unfortunately, bias in salary decisions, whether intentional or not, can lead to employees feeling underappreciated, creating disengagement and eventually increasing attrition rates. This not only lowers morale but also drives up costs for businesses through higher turnover and lost productivity.

To combat this, predicting salaries objectively, without human bias, becomes crucial. By leveraging data-driven models, organizations can determine fair salary ranges based on an individual’s skills, experience, and other measurable factors. This ensures employees are compensated equitably while boosting their sense of value within the organization. **Ultimately, it’s a win-win: happier employees and more profitable, sustainable businesses.**

Objective

Delta Ltd. aims to enhance its employee compensation process by leveraging data-driven insights. The company recognizes the need for fairness, efficiency, and consistency in determining salaries and seeks to eliminate biases or manual discrepancies in its decision-making.

- **Build a Predictive Model**: Utilize historical data to determine the salary to be offered to employees.
- **Minimize Manual Judgments**: Reduce subjectivity in salary decisions by implementing a data-driven approach.
- **Ensure Fairness**: Eliminate discrimination in salaries among employees with similar profiles.
- **Achieve Robustness**: Develop a reliable and scalable solution for consistent salary determination.

Business/ Social Opportunity:

Many organizations have been successful remarkably implementing HR analytics. Here are the few areas I explored:

1. **Improved Compensation and Engagement**:
    - A healthcare organization identified that unequal and highly variable compensation levels led to high attrition rates.
    - Using data analytics, they optimized compensation thresholds, which:
        - Saved over **$100 million**.
        - Enhanced employee engagement and productivity.
        - Reduced attrition rates and overall compensation expenses.

2. **Predictive Behavioural Analytics for Retention**:
    - Another company used predictive analytics to address employee retention:
        - Reduced retention bonuses by **$20 million.**
        - **Halved** employee attrition rates.
        - Identified lack of training and recognition as key drivers of attrition, rather than expensive bonuses, leading to more cost-effective interventions.
3. **McKinsey's Retention Insights**:
    - McKinsey developed machine-learning models to analyse employee behaviour and predict "flight risk."
    - Surprising findings:
        - Lack of mentoring and affiliation were stronger predictors of attrition than performance ratings or compensation.
        - Enhancing mentorship and coaching reduced "flight risk" **by 20–40%.**
    - For consultants participating in training and affiliation programs:
        - Retention likelihood tripled.
        - Resulted in better community engagement and professional development.
4. **Applications Beyond Retention**:
    - Analytics is being extended to other HR areas, such as:
        - Talent acquisition.
        - Performance management.
        - Diversity initiatives.
    - Insights drive strategic decisions tailored to organizational needs.
5. **Social Impact:**
    - **Workplace Equity:** Promotes fair compensation and opportunities by reducing biases in decision-making, fostering a culture of equality and inclusion.
    - **Employee Well-being and Retention:** Improves job satisfaction through mentorship, recognition, and data-driven interventions, reducing attrition and enhancing trust in organizations.
    - **Skill Development and Economic Growth:** Drives targeted training and upskilling initiatives, leading to personal growth, job stability, and broader economic benefits for communities.

Data Collection:

As part of an academic project, we have been provided with a dataset comprising 25,000 job applications to analyze and predict the expected salary of employees based on key attributes.

Data Overview:

- **Shape**: There are 25000 records and 29 columns.
- **Data Types**: There are 16 object data types and 10 int64 and 3 float64 features.
- **Independent & Target Variable**: We need to predict the expected salary of an employee, hence Expected CTC is the target variable and rest are predictors.
- **Check Duplicates**: There are no duplicate records in the database.
- **Check Missing Values:** There are 86853 missing values in the dataset which contributes to 3.4% data
- **Statistical Description:**

Feature	Count	Mean	Std Dev	Min	25th Percentile	Median	75th Percentile	Max
IDX	25000	12500.5	7217.02	1	6250.75	12500.5	18750.25	25000
Applicant_ID	25000	34993.24	14390.27	10000	22563.75	34974.5	47419	60000
Total_Experience	25000	12.49	7.47	0	6	12	19	25
Total_Experience_in_field_applied	25000	6.26	5.82	0	1	5	10	25
Passing_Year_Of_Graduation	18820	2002.19	8.32	1986	1996	2002	2009	2020
Passing_Year_Of_PG	17308	2005.15	9.02	1988	1997	2006	2012	2023
Passing_Year_Of_PHD	13119	2007.4	7.49	1995	2001	2007	2014	2020
Current_CTC	25000	1760945.4	920212.51	0	1027311.5	1802567.5	2443883.25	3999693
No_Of_Companies_Worked	25000	3.48	1.69	0	2	3	5	6
Number_of_Publications	25000	4.09	2.61	0	2	4	6	8
Certifications	25000	0.77	1.2	0	0	0	1	5
International_Degree_Any	25000	0.08	0.27	0	0	0	0	1
Expected_CTC	25000	2250154.5	1160480.14	203744	1306277.5	2252136.5	3051353.75	5599570

Table 1: Statistical Summary of Numerical Features

Feature	Count	Unique	Top	Freq
Department	22222	12	Marketing	2379
Role	24037	24	Others	2248
Industry	24092	11	Training	2237
Organization	24092	16	M	1574
Designation	21871	18	HR	1648
Education	25000	4	PG	6326
Graduation_Specialization	18820	11	Chemistry	1785
University_Grad	18820	13	Bhubaneswar	1510
PG_Specialization	17308	11	Mathematics	1800
University_PG	17308	13	Bhubaneswar	1377
PHD_Specialization	13119	11	Others	1545
University_PHD	13119	13	Kolkata	1069
Current_Location	25000	15	Bangalore	1742
Preferred_location	25000	15	Kanpur	1720
Inhand_Offer	25000	2	N	17418
Last_Appraisal_Rating	24092	5	B	5501

Table 2: Statistical Summary of Categorical Features

- Insights from Statistical Summary:
    - **Experience and Field Experience**: Applicants' total experience averages 12.49 years, with a significant variation (0-25 years). Experience in the applied field has a means of 6.26 years. Higher experience leads to higher compensation.
    - **Educational Background**: The average passing year for graduation is 2002, with a post-graduation average year of 2005. Advanced education typically leads to higher compensation.
    - **Current and Expected CTC**: The average current CTC is 1.76 million, with the expected CTC averaging 2.25 million. By understanding current compensation and the expectations of candidates, HR can align their offers to market standards and reduce the risk of losing top talent due to salary mismatches.
    - **Education and Specializations:** Attributes like PG education and fields like Mathematics and Chemistry significantly influence salary predictions.
    - **Industry and Roles:** Dominant industries (e.g., Training) and roles (e.g., HR) provide benchmarks for competitive salary offerings.
    - **Location Preferences and Offers**: Insights from preferred locations and in-hand offers help HR tailor strategies to attract top talent effectively.

Data Dictionary:

Attribute	Description
Applicant_ID	Unique identifier for each applicant
Total_Experience	Total years of experience the applicant has accumulated
Total_Experience_in_field_applied	Years of experience specifically in the field applied for
Department	The department in which the applicant is being considered (e.g., HR, Banking)
Role	The applicant's current role or the role they are applying for
Industry	The industry in which the applicant has experience (e.g., Analytics, Insurance)
Organization	Name of the organization the applicant has worked in
Designation	Job designation or title of the applicant
Education	The highest level of education achieved (e.g., PG, Grad)
Graduation_Specialization	The specialization in the undergraduate education
University_Grad	University from where the applicant graduated (undergraduate)
Passing_Year_Of_Graduation	Year the applicant graduated from undergraduate studies
PG_Specialization	The specialization in postgraduate education
University_PG	University from which the applicant received their post-graduate degree
Passing_Year_Of_PG	Year of post-graduate completion
PHD_Specialization	Specialization area during the Ph.D. if applicable
University_PHD	University from which the applicant completed their Ph.D.
Passing_Year_Of_PHD	Year of Ph.D. completion
Current_Location	Location where the applicant is currently based
Preferred_location	The location where the applicant would prefer to work
Current_CTC	Current Cost to Company (CTC) for the applicant
Inhand_Offer	Indicates whether the applicant has an in-hand offer (Y/N)
Last_Appraisal_Rating	The rating given during the last performance appraisal
No_Of_Companies_worked	The number of companies the applicant has worked for
Number_of_Publications	Number of publications authored by the applicant
Certifications	Number of certifications the applicant holds
International_degree_any	Whether the applicant holds an international degree (1 = Yes, 0 = No)
Expected_CTC	The expected salary offered to the applicant

Table 3: Data Dictionary

Exploratory Data Analysis:

Univariate Analysis: Numerical Variables

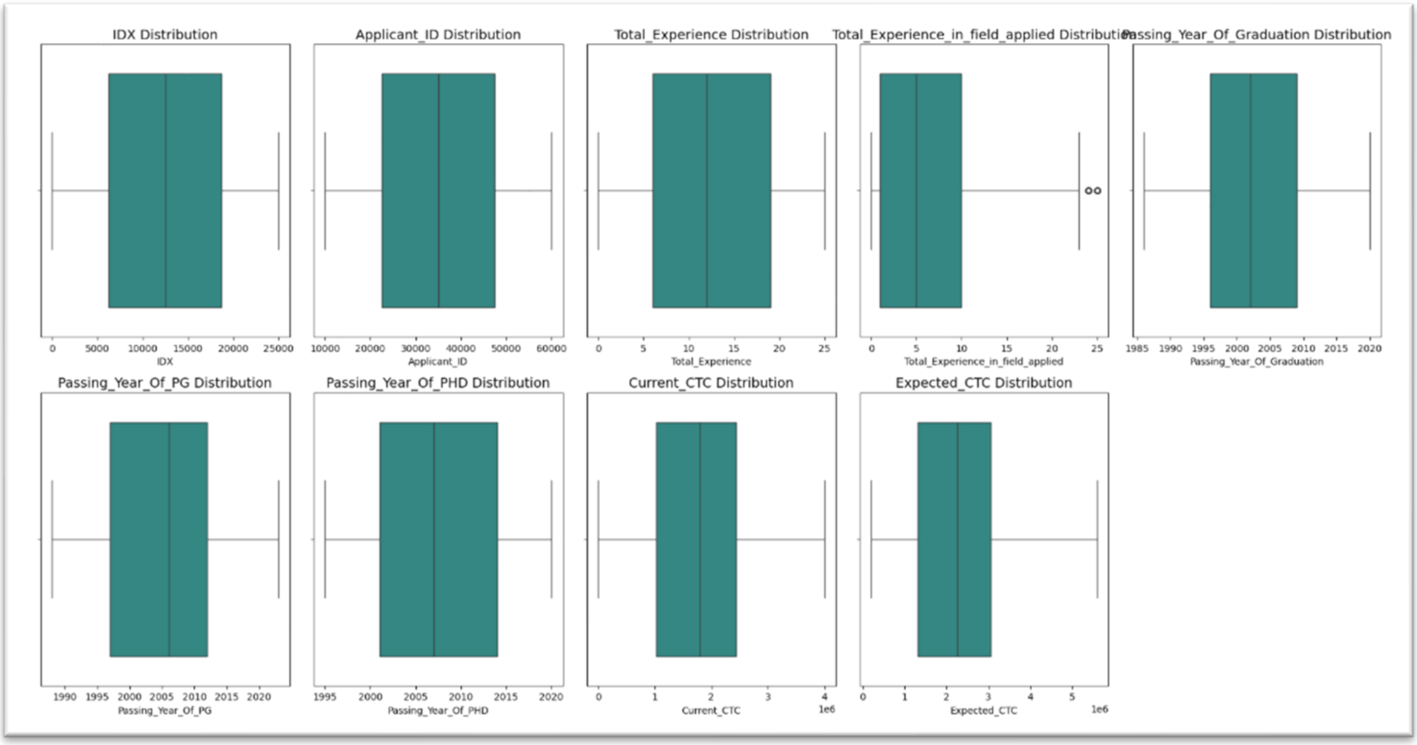

Fig 4: Box Plot of Numerical Features

Observations:

- ✓ IDX is a unique sequential number which does not contribute to the data analysis. This can be removed.
- ✓ Application ID is a unique identifier for each application of all the positions the organization has vacancies. This also can be removed.
- ✓ Total Experience goes till 25 starting from freshers which indicates there are diverse opportunities for senior top management and those starting the career.
- ✓ 75% of applicants have at most 10 years of experience in the field they have applied for, However, there are people who have 25 years of experience in the field itself.
- ✓ Applicants with advanced education and earlier graduation years (e.g., in the 1990s) are likely to bring substantial professional experience, which increases their potential value to the organization and justifies higher compensation packages.
- ✓ Currently CTC and Expected CTC are mostly following the same pattern, however, based on the location, upskills and market standards it can vary significantly.

Univariate Analysis: Categorical Variables

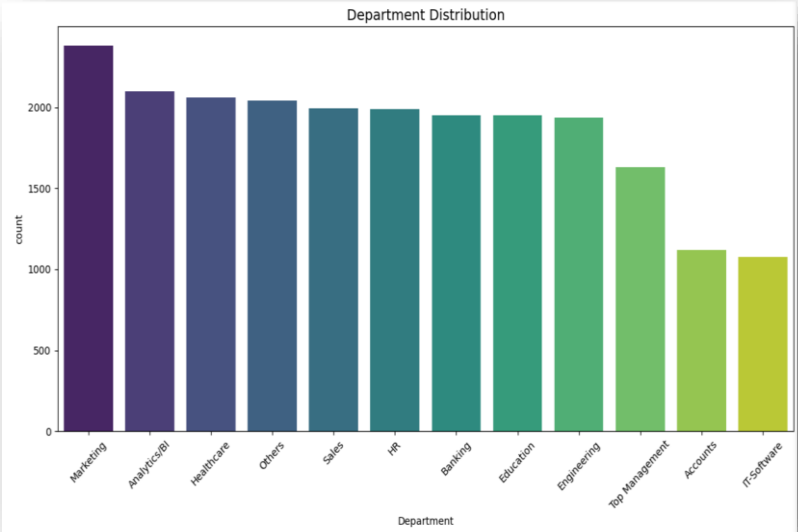

Fig 5: Count plot of Department

Observation:

- ✓ The Marketing department has the highest representation, indicating its prominent role in the dataset.
- ✓ **Analytics/BI and Healthcare are significant contributors**: These departments also have a high frequency, reflecting growing industry demand for analytical and healthcare expertise.
- ✓ **IT-Software and Accounts are less represented**: These departments show the lowest counts, suggesting they are either niche roles in the dataset or not primary recruitment focus areas.
- ✓ Departments such as Sales, HR, Banking, Education, and Engineering are evenly distributed, indicating diverse hiring trends across industries.

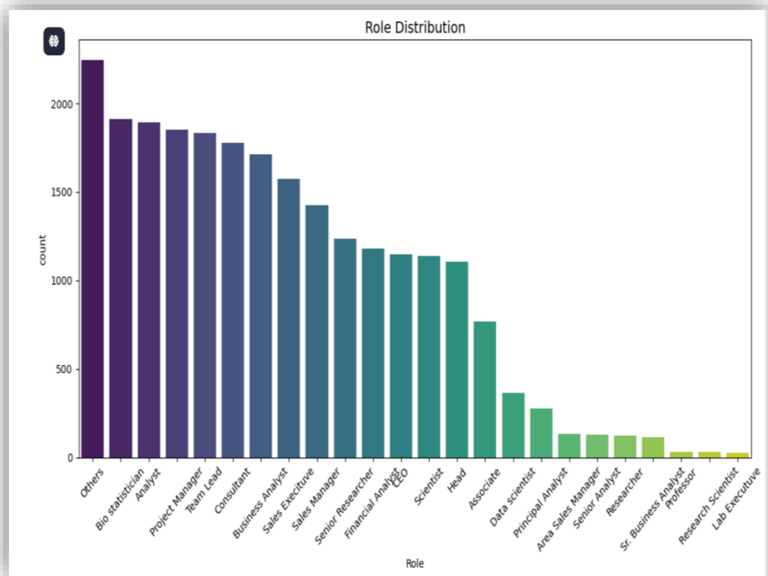

Fig 6: Count plot of Role

Observation:

- ✓ Role Diversity: The dataset captures applicants in a wide variety of roles, with "Others," "Biostatistician," "Analyst," and "Project Manager" being the most represented roles.
- ✓ Roles such as "Lab Executive," "Research Scientist," and "Professor" are significantly underrepresented compared to others.
- ✓ The dominance of roles such as "Team Lead," "Consultant," and "Sales Executive" suggests a focus on mid-level management and operational roles in the dataset.
- ✓ Identifying demand for roles can help the organization prioritize recruitment strategies and allocate resources based on applicant trends and business needs.

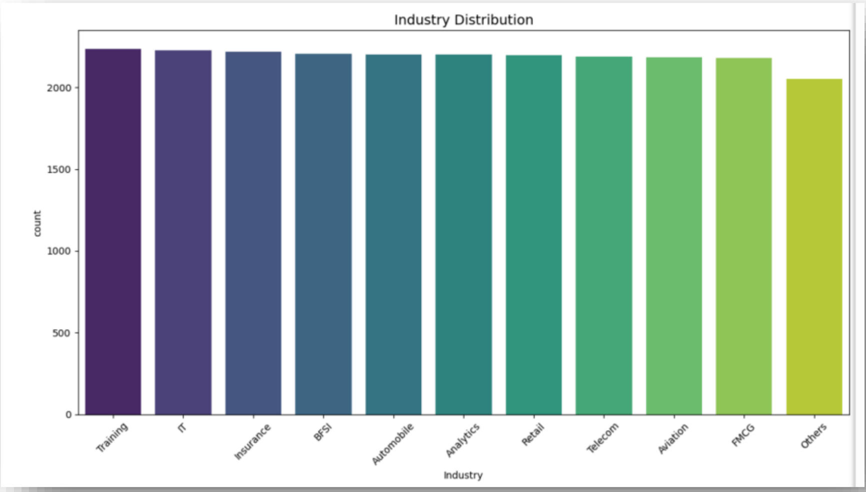

Fig 7: Count plot of Industry

Observation:

- ✓ The applicants are evenly distributed across multiple industries such as Training, IT, Insurance, BFSI, Automobile, Analytics, and more.
- ✓ Industries like FMCG and Aviation show significant representation, reflecting a diverse applicant pool across key economic sectors.

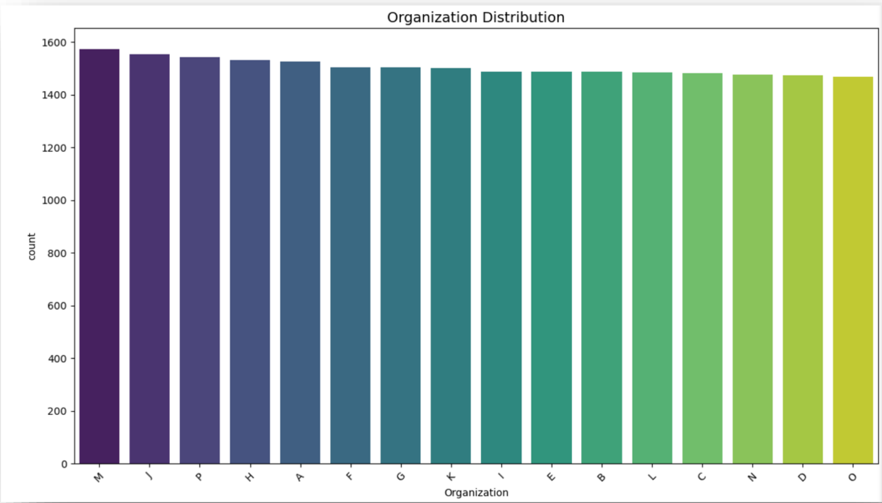

Fig 8: Count plot of Organization

Observation:

- ✓ Organization data is not intuitive; however, we see that there is an equal distribution of the applicants from all the available organizations.

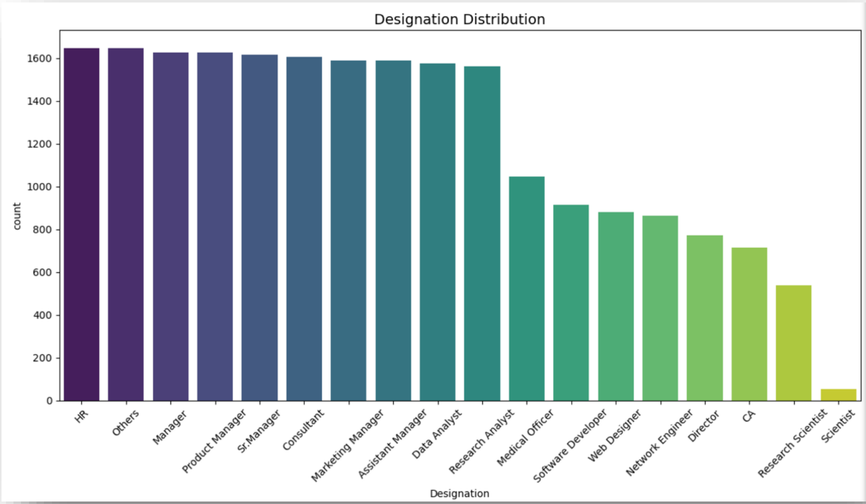

Fig 9: Count plot of Designation

Observation:

- ✓ HR, Managers, and Product Managers may represent the backbone of operations or administration.
- ✓ Designations like Medical Officers, Software Developers, and Data Analysts have medium representation, these positions might require specialized skills and could be directly tied to core operations.
- ✓ Positions like Research Scientist and Scientist are the least represented, R & D may not have enough scope in the organization.
- ✓ There is a reasonable mix of Senior Management Roles (Manager, Director) and Operational Roles (Developer, Analyst) which indicates smooth functioning of the organization and clear career pathway.

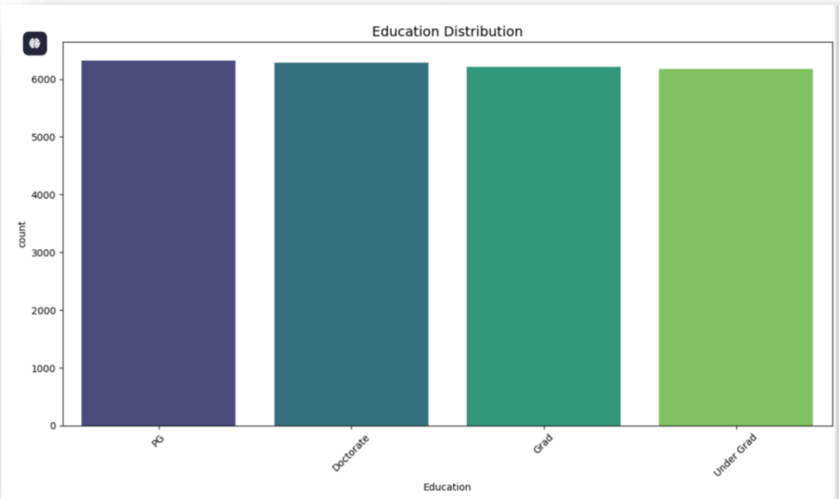

Fig 10: Count plot of Education

Observation:

The applicants are evenly distributed across all 4 educational level.

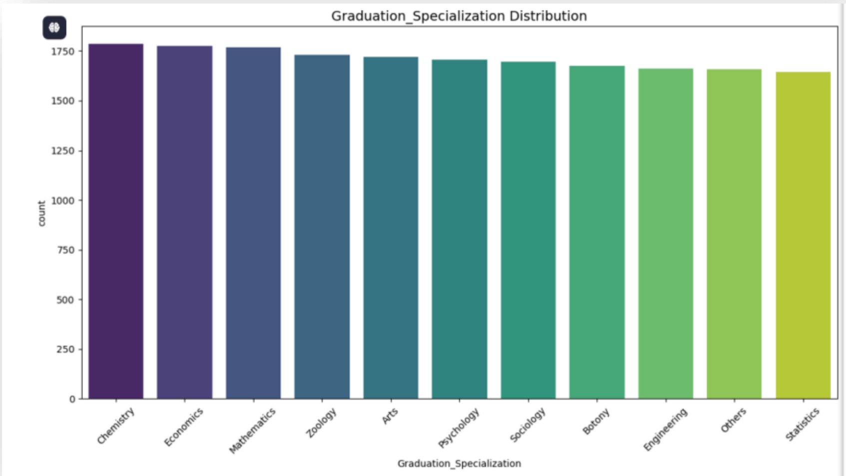

Fig 11: Count plot of Graduation Specialization

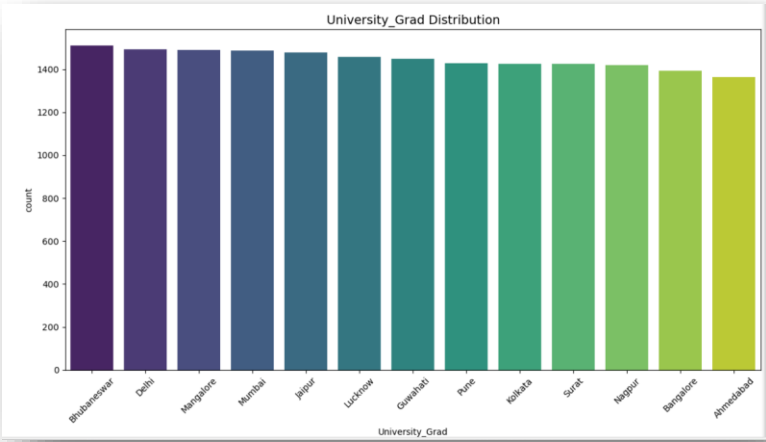

Fig 12: Count plot of Under Grad City

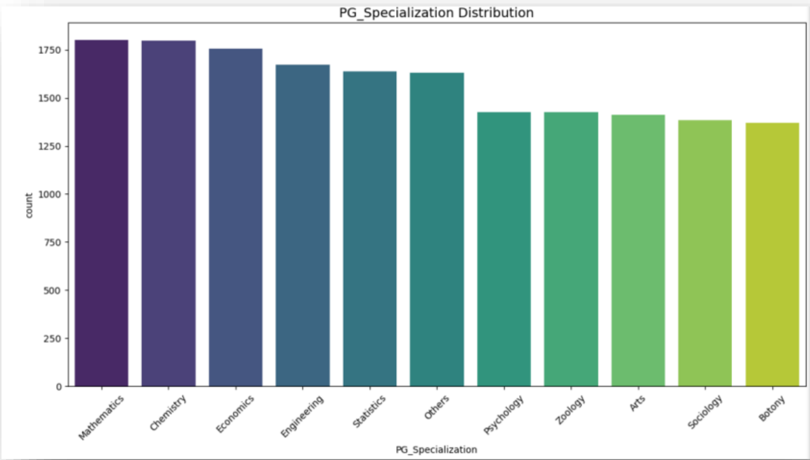

Fig 13: Count plot of PG Specialization

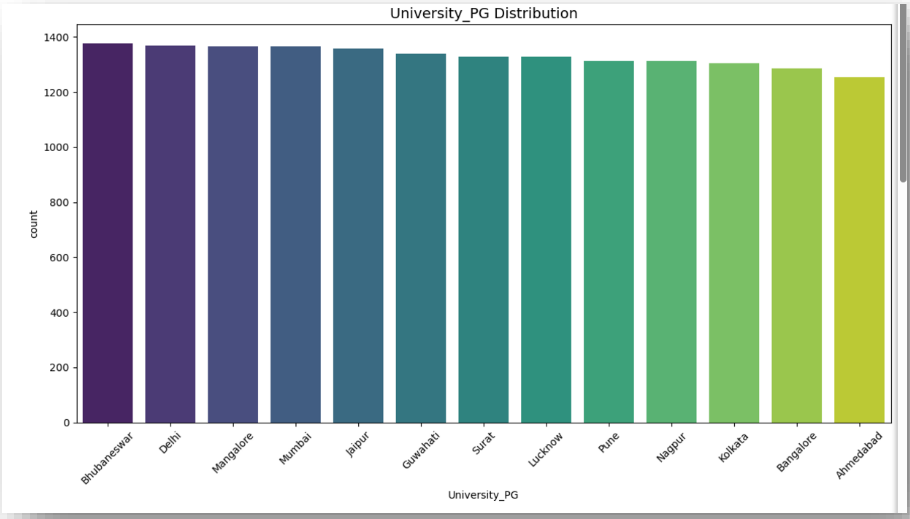

Fig 14: Count plot of PG City

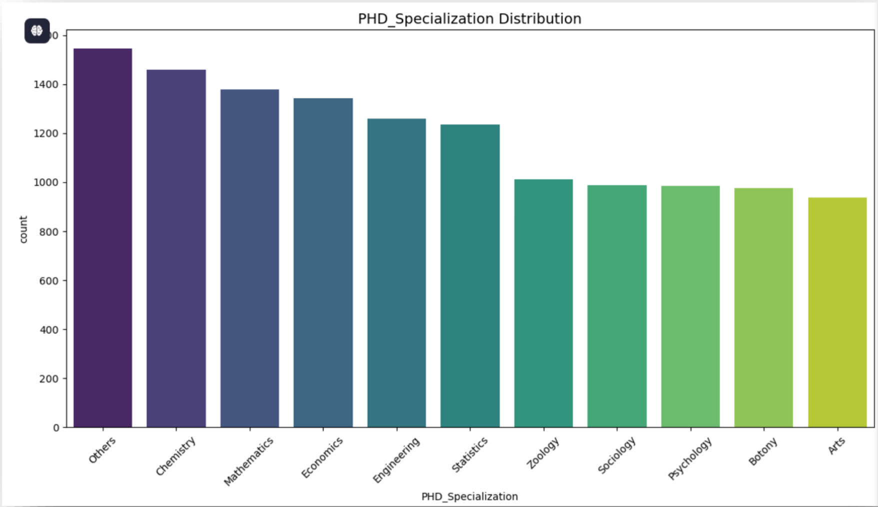

Fig 15: Count plot of PHD Specialization

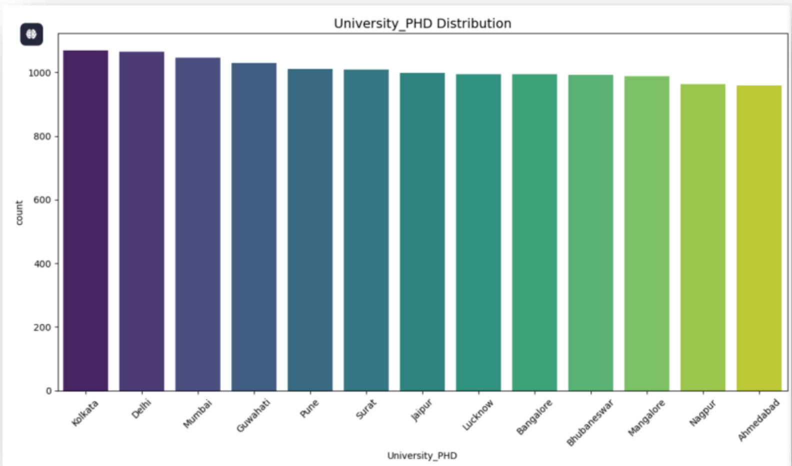

Fig 16: Count plot of PHD City

Observation:

- ✓ There is a significant number of applicants coming from Chemistry, Economics and Mathematics backgrounds in all different education levels (Grad, PG, PHD).
- ✓ There is no major distinction of the cities they passed out from across all levels.

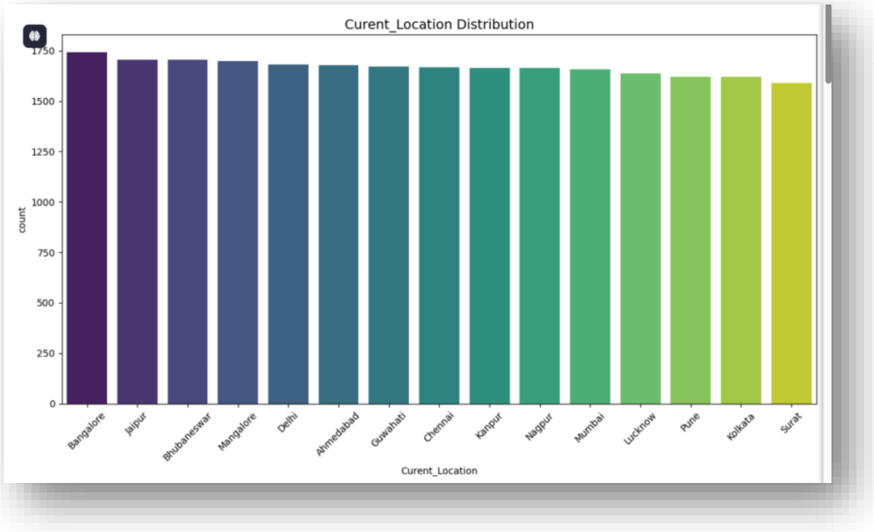

Fig 17: Count plot of Current Location

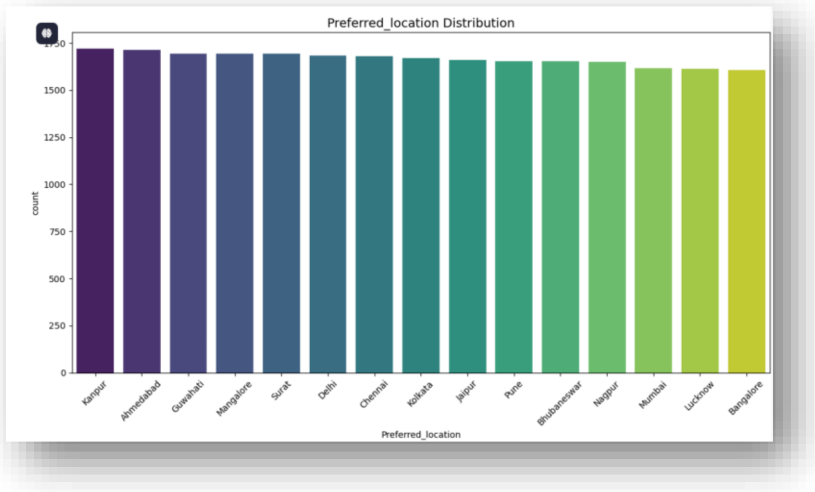

Fig 18: Count plot of Preferred Location

Observation:

- ✓ Though a huge number of applicants are based in Bangalore, it seems that they don't prefer the city when it comes to job change. Analysis needs to be to understand the underlying reason.
- ✓ Kanpur, Ahmedabad, Guwahati are the most preferrable cities applicants are looking for.

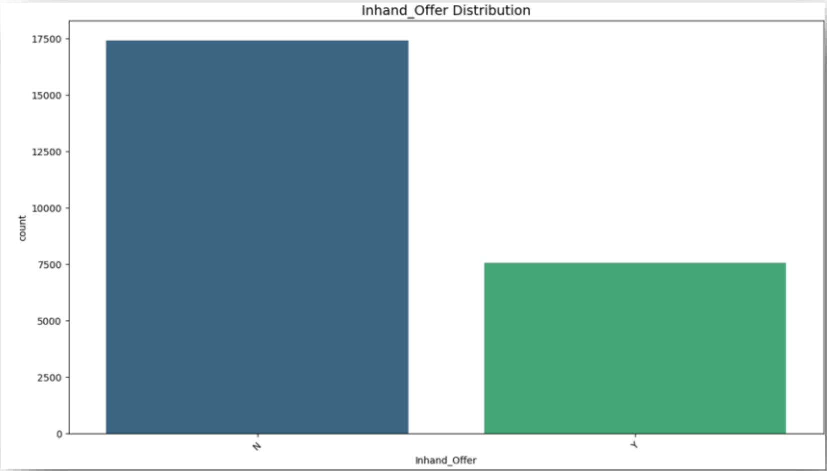

Fig 19: Count plot of Inhand Offer

Observation:

70% of applicants don't have any offer at hand when applying for job.

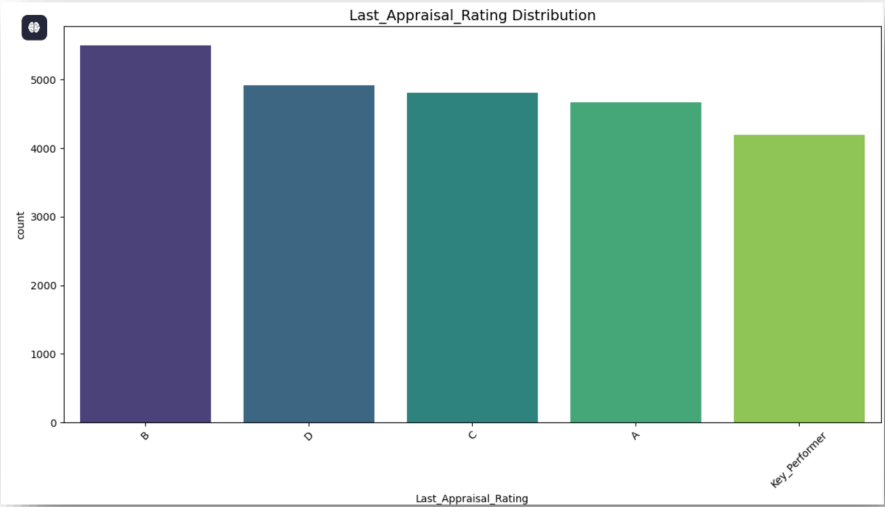

Fig 20: Count plot of Last Appraisal Rating

Observation:

- ✓ B has the highest count, suggesting that a significant proportion of applicants are average or slightly above average.
- ✓ D, C & A are mostly equally distributed.
- ✓ A few applicants are high performers.

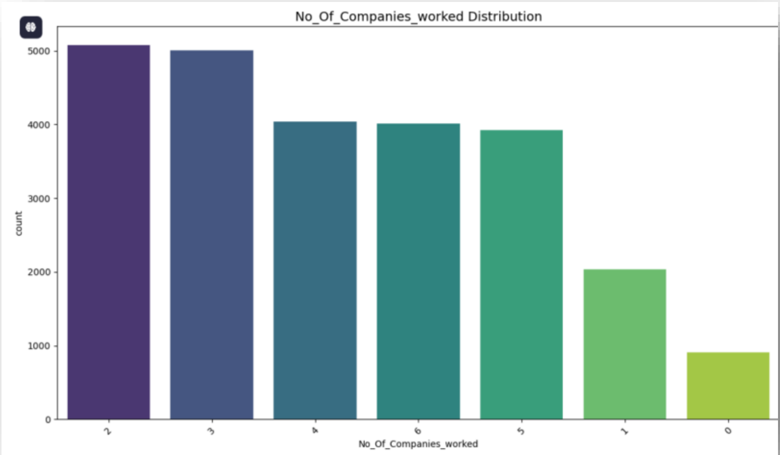

Fig 21: Count plot No of Companies Worked

Observations:

- ✓ Applicants with experience from 2-5 companies may bring sufficient professional experience without appearing overly transient, which may be attractive to the organization.
- ✓ The relatively low number of freshers (0 companies) presents an opportunity to implement programs targeting early-career talent to build a long-term pipeline.
- ✓ Employees with experience at more than 5 companies may be seen as job hoppers. For such applicants, focus on understanding their reasons for switching roles during interviews or assessments to gauge their stability.

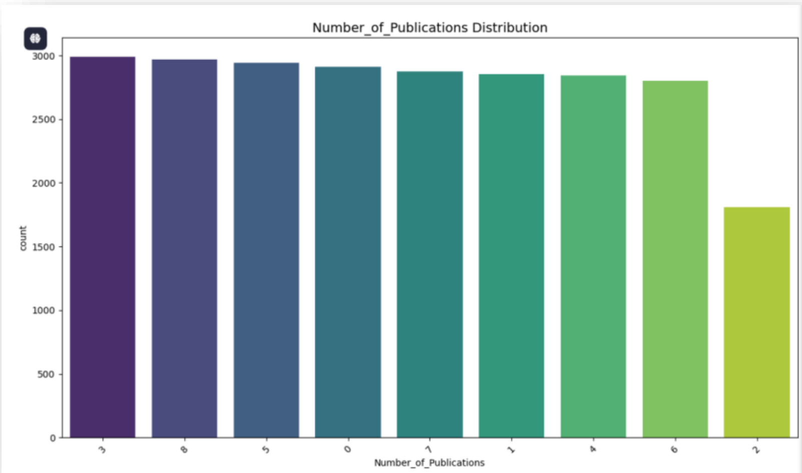

Fig 22: Count plot of No of Publications

Observation:

- The maximum number of applicants has 3 publications, but difference is not much significant to draw any conclusions.

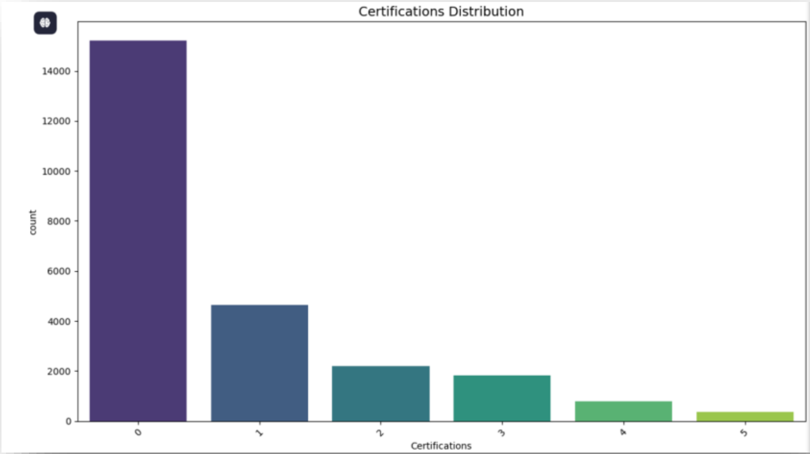

Fig 23: Count plot of Certifications

Observation:

- Most of the applicants don’t hold any certification, indicating they are mostly coming from direct educational institutions and real work experience.
- This also can be an indication of lack of upskilling.

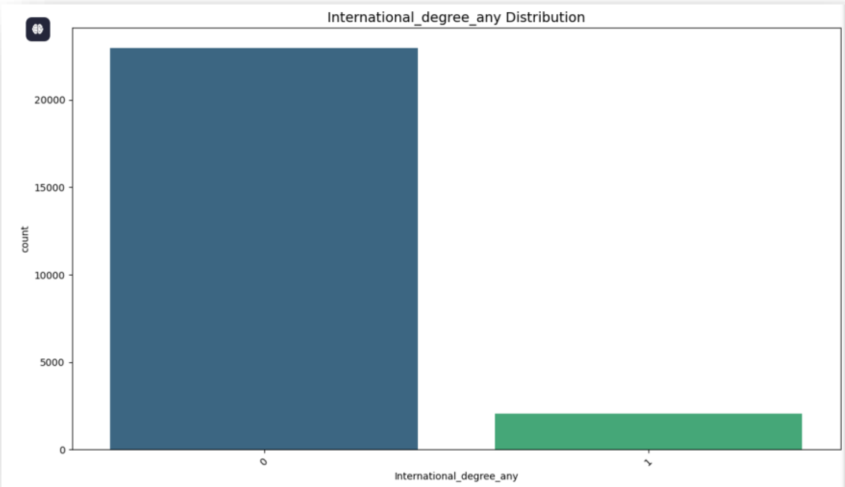

Fig 24: Count plot of International Degree

Observation:

- Most applicants lack an international degree, and those who possess such a degree tend to work abroad.

Bivariate Analysis:

Categorical Variables Vs Target

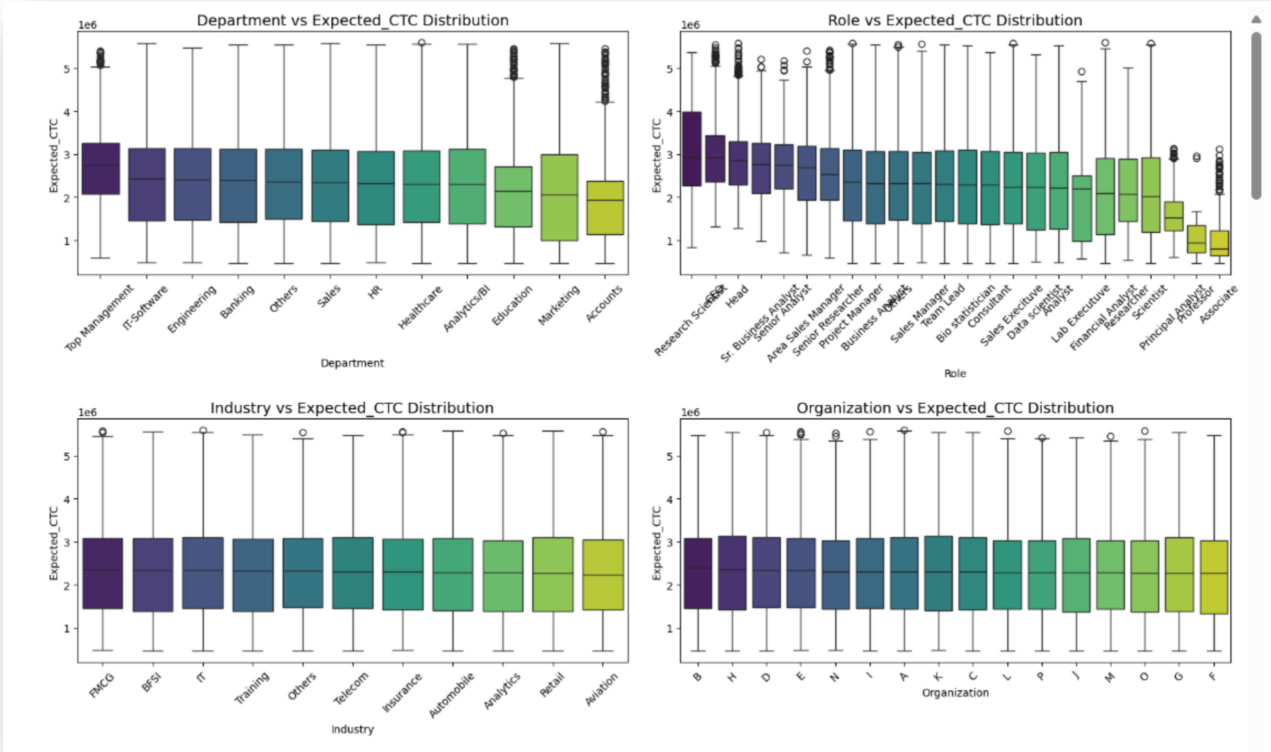

Fig 25: Box Plot of Categorical vs Target

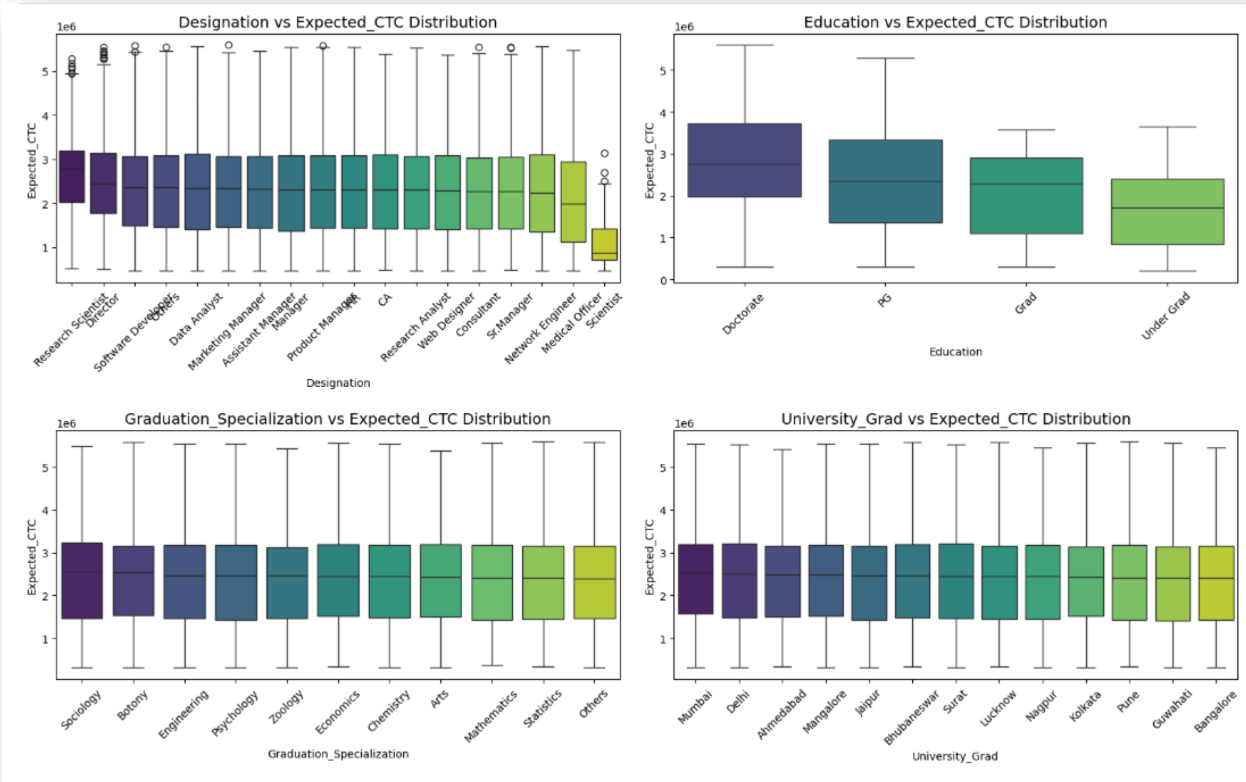

Fig 26: Box Plot of Categorical vs Target

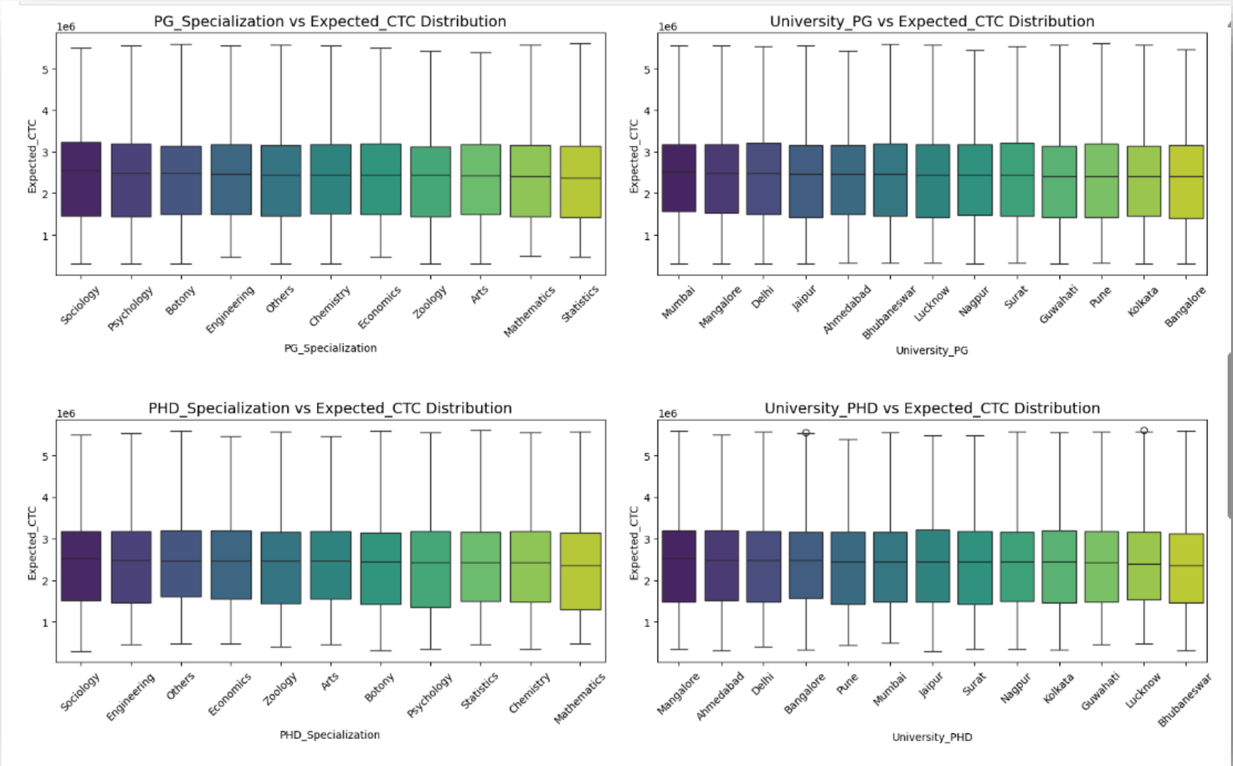

Fig 27: Box Plot of Categorical vs Target

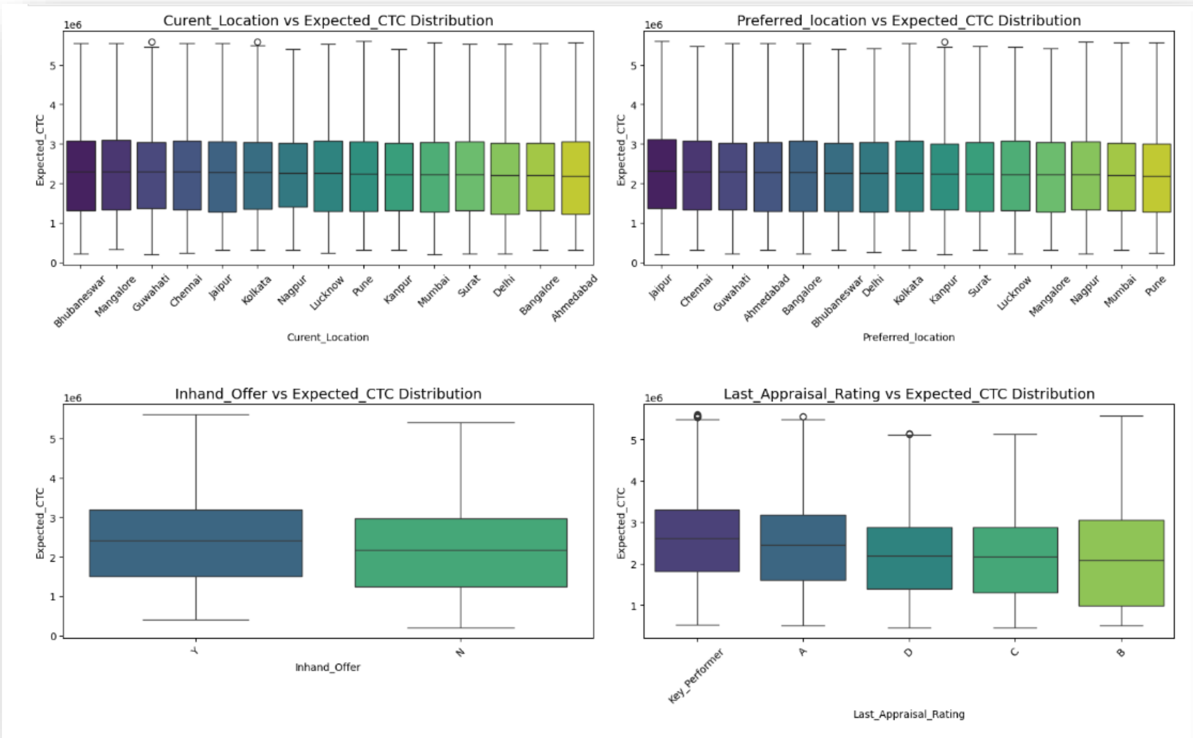

Fig 28: Box Plot of Categorical vs Target

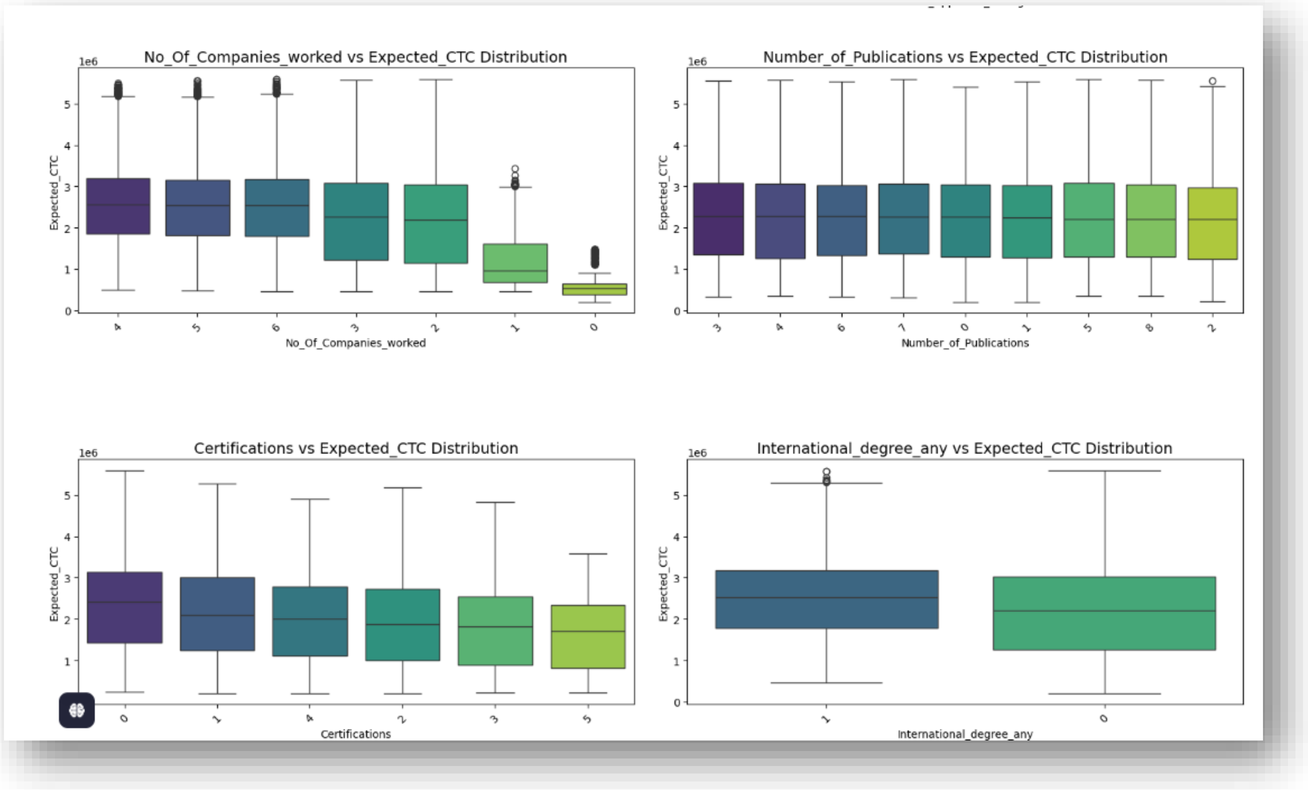

Fig 29: Box Plot of Categorical vs Target

Observation:

- Departments such as **Top Management, IT Software, Banking, and Engineering** offer **higher pay**, reflecting their critical contribution to organizational profitability and strategic objectives. In contrast, the **Accounts department**, while essential for operational stability, tends to have comparatively **lower pay** due to its supportive nature and limited direct impact on revenue generation or innovation with a few outliers.
- Roles such as **Research Scientist, Head, Senior Business Analyst, Senior Analyst, and Area Sales Manage**r command **higher salaries** due to their critical contributions to innovation, business growth, and revenue generation. Conversely, roles like **Associate (entry-level positions), Professor, and Principal Analyst** tend to have **lower pay**, reflecting either their entry-level status, academic focus, or limited direct impact on organizational profitability.
- **Role and Designation** are having similar trends when it comes to the salary, however it is surprising that scientists tend to have comparatively minimal salaries which might indicate they are very new to industry and doesn't have significant Contribution in revenue generation.
- Very intuitively the professionals having higher education level tend to have higher payroll, similar patterns to the applicants having offer at hand and high performer.
- The more companies an individual has worked for, the higher their professional expertise and adaptability. This experience translates into greater value for organizations, often reflected in higher compensation.
- The features like **Industry, Organization, Current and Preferred Location, Education related specifications (Passing Year, Specialization and University)** each

individually don't have much impact on the expected salary, need further analysis if combination of any 2/3 has an impact or not.
- Having an international degree is always a bonus very certainly.
- No of certifications an applicant has, doesn't seem to be a strong predictor of compensation.

Bivariate Analysis: **Numerical Variables Vs Target**

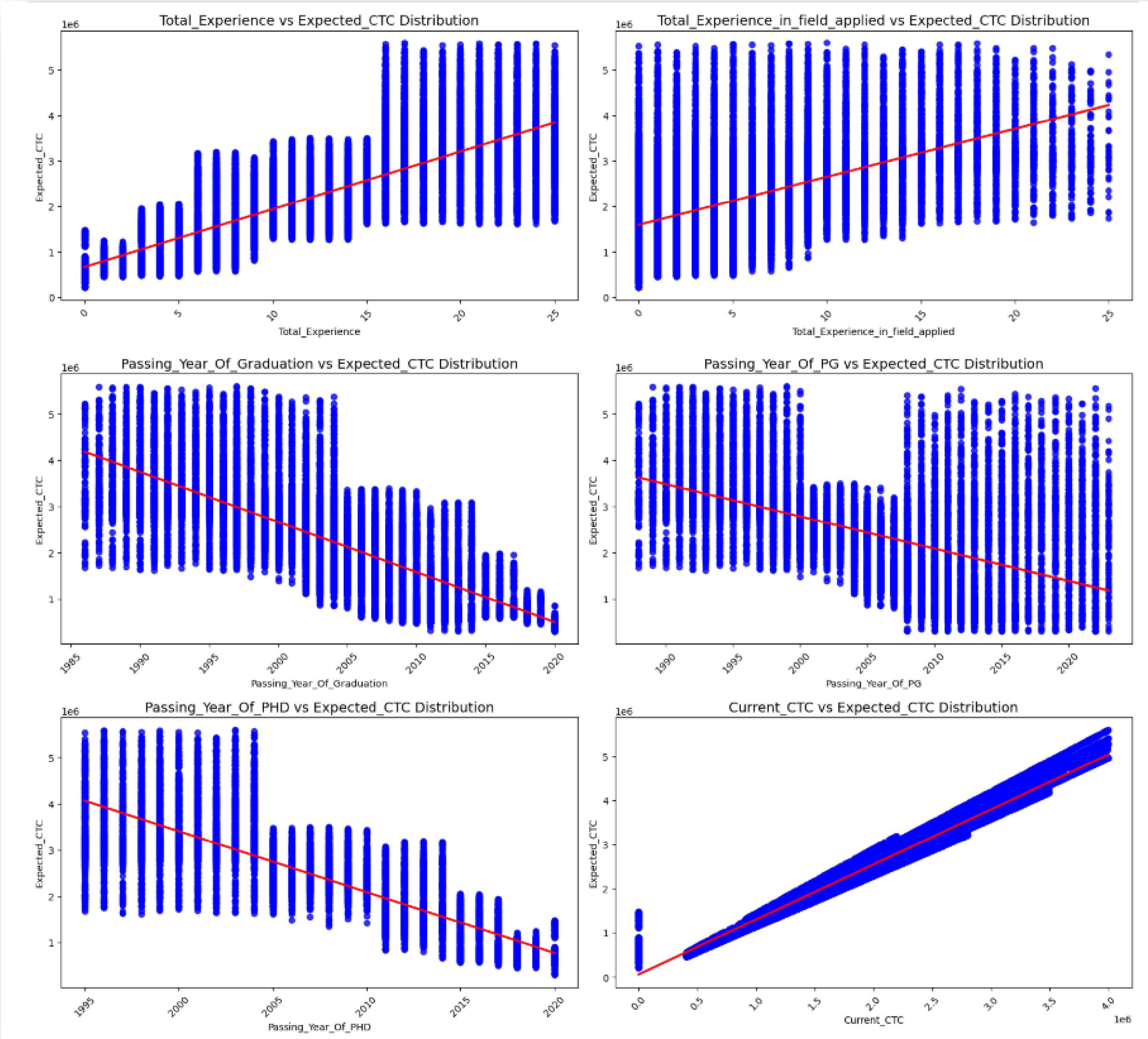

Fig 30: Scatter Plot of Numerical Variables vs Target

Observations:

- Increased Experience Leads to Higher Compensation
- Seniority Reflects Expertise and Justifies Higher Salary
- Expected Salary Generally Aligns with Current CTC, with Some Variations.

Multivariate Analysis:

Let's check how the numerical features are correlated to each other.

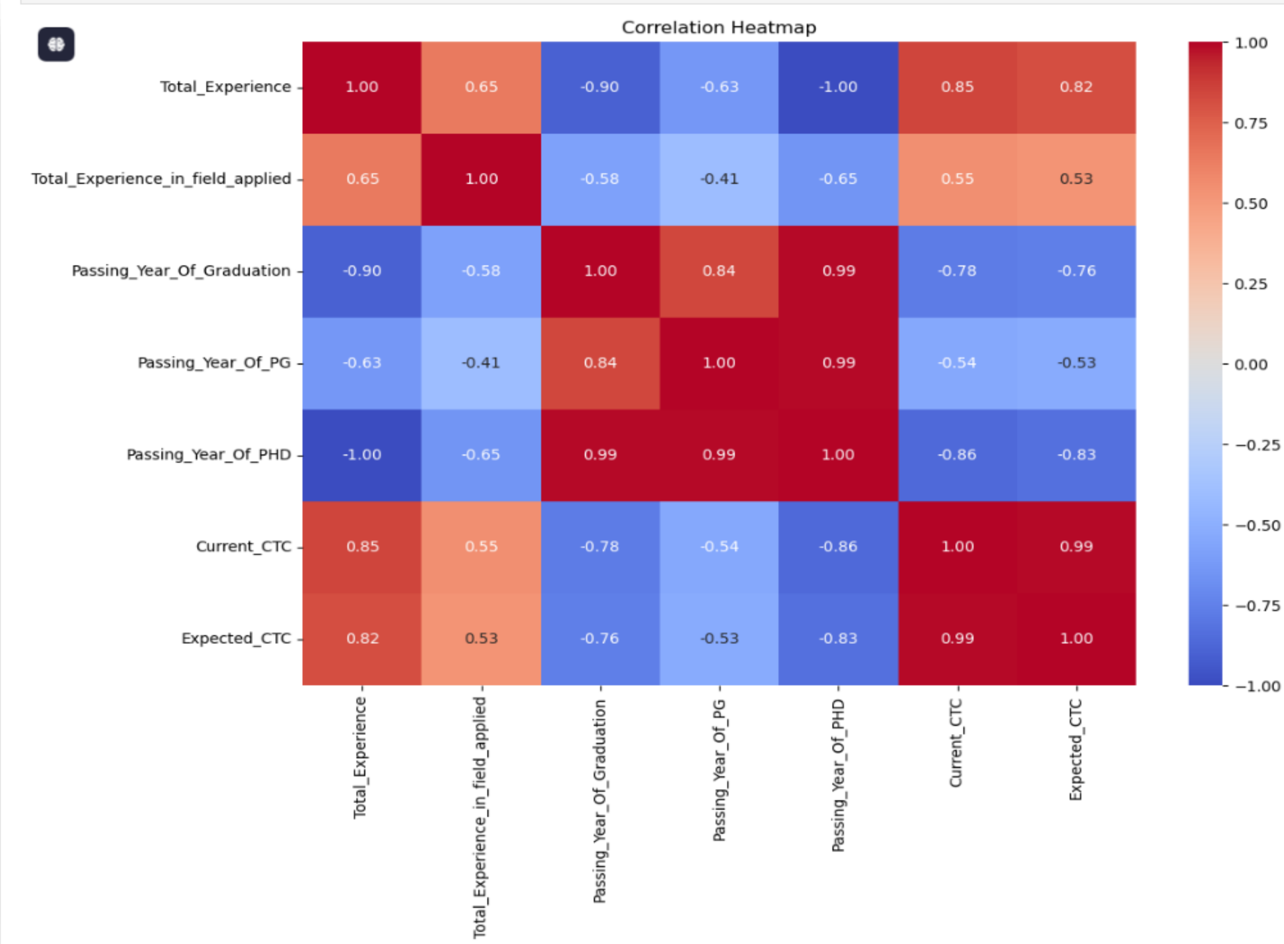

Fig 31: Heat Map

Observations:

- There is a high correlation between the total experience and compensation, a relatively less in the field itself the applicant has applied.
- The passing year of PHD has a relatively higher correlation with salary compared to other education levels.
- There is 99% correlation between current and expected CTC

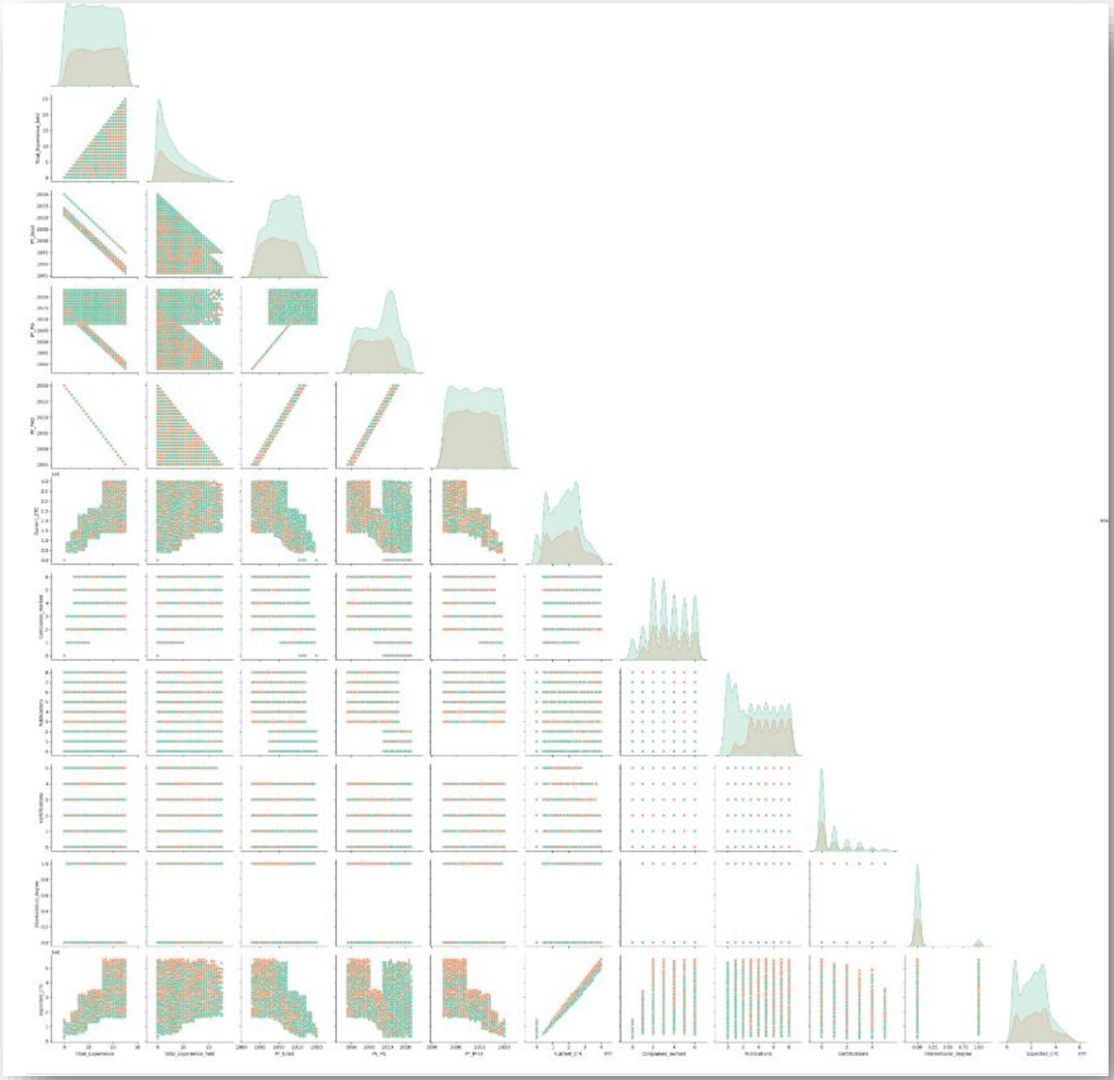

Fig 32: Pair Plot

Check the Significance of Features on Target:

- From Fig 30,31 we see that there is a strong linear relationship between Current CTC and Expected CTC, let's perform Hypothesis testing to check if there is any significant Difference between these 2 else we can remove Current CTC.

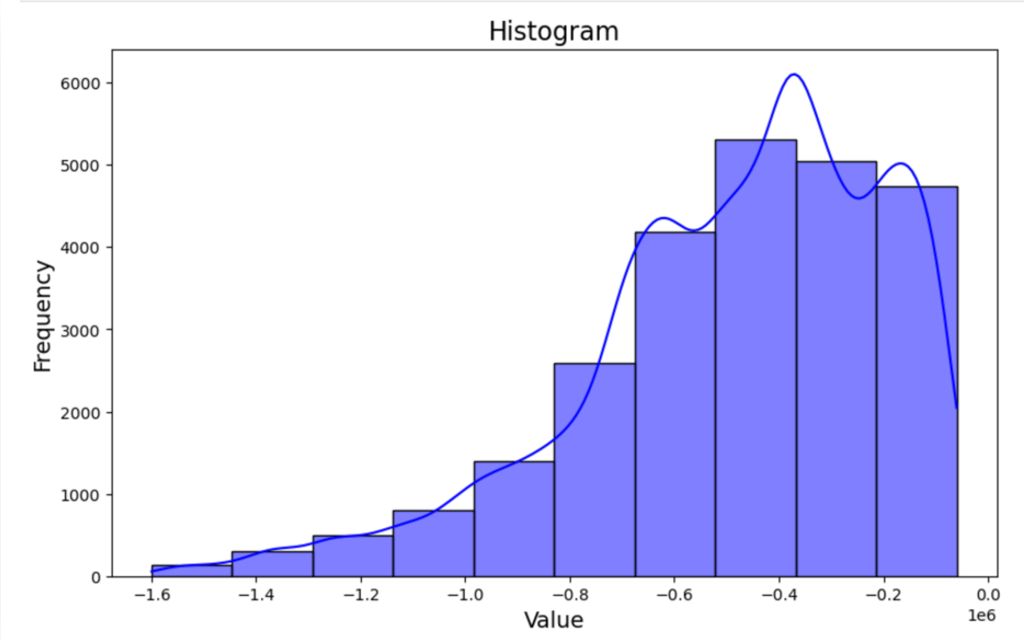

Fig 33: Histogram of Difference b/w Current & Expected CTC

✓ We performed Shapiro test to check if the difference is normally distributed. As seen from the above graph and from the test (p-value<0.05), difference is right skewed.

✓ We performed the Wilcoxon test to check if this difference is significant or not. The result shows the difference is statistically significant (p-value<0.05)

- From the bivariate analysis of categorical features vs target, we saw that there are few features (Industry, Organization) which don't have much impact on the Expected CTC. Let's confirm this assumption statistically.

Category	p-value
Department	0.00
Role	0.00
Industry	0.18
Organization	0.28
Designation	0.00
Education	0.00
Graduation Spec	0.75
University Grad	0.84
PG Spec	0.55
University PG	0.87
PHD Spec	0.13
University PHD	0.92
Current Location	0.69

Preferred Location	0.65
Inhand Offer	0.00
Last Appraisal Rating	0.00
No of Companies Worked	0.00
No of Publications	0.15
Certifications	0.00
International Degree	0.00

Table 3: p-values from Anova

- ✓ We see that industry, organization, education details (UG, PG, PHD), Location Details, No of Publications don't have statistical significance (p-value>0.05) on Expected CTC individually.
- ✓ After going through a couple of pair wise Anova testing, we can conclude that UG, PG details (except Passing Year-significant missing values), Preferred Location are statistically significant to predict expected salary.

Removal of Unwanted Variables:

- The variables we are going to remove **IDX, Applicant ID** (as they do not hold useful information), **Industry, Organization** (not statistically significant to predict the expected salary), **PHD Specialization, Passing Year of PHD, University PG, Passing Year of Grad, Passing Year of PG** (having significant missing values).
- There are 14 records for which Department, Role and Designation are set Others which don't contribute to bringing insight into the analysis in the context of 25000 records. Hence removing all 14 records.
- There are 24986 applicants' data with 20 features after cleaning.

Data Pre-processing:

Missing Value Treatment:

PHD_Specialization	47.524
University_PHD	47.524
Passing_Year_Of_PHD	47.524
PG_Specialization	30.768
University_PG	30.768
Passing_Year_Of_PG	30.768
Graduation_Specialization	24.720
University_Grad	24.720
Passing_Year_Of_Graduation	24.720
Designation	12.516
Department	11.112
Role	3.852
Industry	3.632
Organization	3.632
Last_Appraisal_Rating	3.632

Fig 34: Missing Value % for each feature

- As the missing values of PHD Specialization, University PHD, Passing_Year_Of_PHD, Passing Year of Grad and Passing Year of PG features contribute to significant missing data in the data set, we have removed those features.
- We have done KNN Imputation to handle other missing values after performing scaling in the entire data set.

Outlier Treatment:

There are no outliers in the dataset as we observed in the univariate analysis.

Variable Transformation:

- There are 16 categorical variables which need to be encoded for KNN Imputation of missing values and model building at later point of time.
- Internation Degree, Certifications, Publications and No of Companies worked these features (though categorical) hold numeric values. Hence just changing the data type back to int (data type was initially converted from int to object for visualization purpose)
- We performed Ordinal encoding for the rest of the categorical features.
- We performed scaling (Min Max Scaling) on the numerical data and combined with the categorical ones.
- KNN imputation has been performed with 3 nearest neighbors to remove the missing value in the final data set.

Addition of New Variable: No variable needs to be added here

Check Data Imbalance:

As part of univariate analysis, we observed that the target variable Expected CTC is evenly distributed in the dataset, and there are no outliers. Hence data is balanced.

Check Clustering:

- Clustering is required to see the pattern of the applications in the entire data set based on the roles, departments, degrees and the location and most importantly salary.
- We performed k-means clustering experimented with an iteration of number of clusters from 2 to 12 and found out the silhouette score to determine the optimum number of clusters.
- We come to conclusion of 2 clusters based on the experiment.

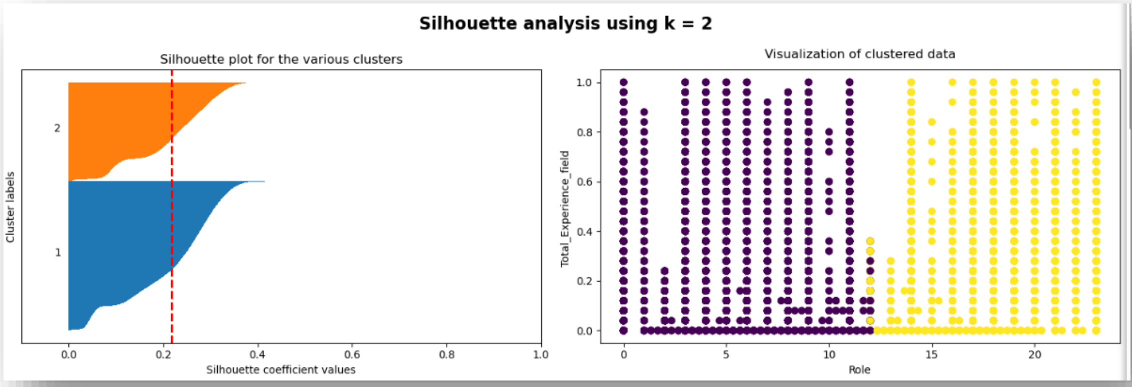

Fig 35: 2 Clusters Visual

Cluster 0:

- Slightly lower international degree (0.091) and lower Expected Salary (0.376) than Cluster 1.
- Represents candidates with moderate work experience and relatively balanced characteristics across features.

Cluster 1:

- Higher Expected Salary (0.382) and more diverse Role (18.33) values.
- Represents candidates with higher diversity in roles, potentially higher experience levels, or higher salary expectations.

Key Insights:

1. **Cluster 0:**
   Profile: Moderate experience, certifications, and education level.
   Expected Salary: Slightly lower compared to Cluster 1.
   Current Location: Candidates in this cluster may be more localized or prefer stability.
2. **Cluster 1**:
   Profile: Likely more experienced, with higher certifications and diverse roles.
   Expected Salary: Higher than Cluster 0.
   Career Progression: This cluster might include candidates with a stronger career trajectory or ambitions.
3. **Key Predictors:** Total experience, advanced education, and current salary strongly influence expected salary.
4. **Low Impact Features:** Industry, organization, and location preferences have minimal effect on salary predictions.
5. **High-Paying Roles:** Senior and specialized roles like Research Scientist and Team Lead demand higher salaries.